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EIGSEP Battery Configuration and Setup

Richard Rodriguez[1, 2]

[1] *Radio Astronomy Laboratory, University of California Berkeley, Berkeley, CA 94720, USA*
[2] *Department of Physics and Astronomy, San Francisco State University, San Francisco, CA 94132, USA*

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1. INTRODUCTION

This document describes how to build, wire, and connect the main battery pack used for EIGSEP. The pack consists of two independent modules: a 4S (12V) module and a 2S (6V) module, each built from $LiFePO_4$ cells. Each module has its own Battery Management System (BMS) for cell protection and voltage balancing, its own inline fuse holder for overcurrent protection, and its own TX1 WH2 circuit breaker for manual disconnect. Together the two modules power the entire EIGSEP system and weigh approximately 50 lbs (∼23 kg).

A reader with no prior experience building battery packs should be able to follow this document from start to finish. Each section explains not just *what* to do, but *why*.

2. BACKGROUND

2.1. *$LiFePO_4$ cells*

$LiFePO_4$ (lithium iron phosphate) cells have a nominal voltage of 3.2 V each. Connecting cells in series (positive terminal of one cell to negative terminal of the next) adds their voltages. Four cells in series (4S) gives $4 \times 3.2 = 12.8$ V. Two cells in series (2S) gives $2 \times 3.2 = 6.4$ V.

2.2. *The Battery Management System (BMS)*

A BMS protects the cells from damage by monitoring each cell's voltage and cutting off current if any cell goes outside safe limits (overcharge, overdischarge, or overcurrent). It connects to the battery on two power terminals:

- `B-` — the main battery negative input to the BMS.
- `P-` — the protected negative output from the BMS to the load.

The BMS sits in the *negative* path only. The positive wire (`B+`) bypasses the BMS entirely and goes directly to the fuse and breaker. When the BMS trips, it opens the `P-` output, breaking the circuit and stopping current flow.

The BMS also has a separate low-current *balance port* (sometimes called the sampling connector). This multi-pin connector carries one wire per cell junction so the BMS can measure each cell's individual voltage. It carries no load current and must be connected before the main power wires.

2.3. *Fuse and circuit breaker*

Each module uses two layers of overcurrent protection on the positive line:

- **Inline fuse holder** — an ATO/ATC blade fuse holder wired close to the battery positive terminal. Each module has its own dedicated fuse holder. Protects the wiring between the battery and the breaker from a catastrophic short circuit. Sacrificial: the fuse must be replaced after blowing.
- **TX1 WH2 circuit breaker** — a hydraulic-magnetic breaker downstream of the fuse. Acts as a manual on/off switch and also trips automatically under sustained overcurrent. Resets by flipping the handle back on — no replacement needed.

2.4. *Output connector*

Each module's output is a single N-type coaxial connector. The center pin carries the positive supply (`B+` path), and the outer shield carries the negative return (`P-` path). One connector provides a complete circuit to the load.

3. ASSEMBLY

3.1. *Building the 4S (12V) module*

Connect four 3.2 V $LiFePO_4$ cells in series using shunts: the negative terminal of cell 1 connects to the positive terminal of cell 2, the negative of cell 2 to the positive of cell 3, and the negative of cell 3 to the positive of cell 4 (Figure 2). The main positive terminal of the pack (`B+`) is the free positive of cell 1. The main negative terminal (`B-`) is the free negative of cell 4, which connects to the BMS. Arrange the cells so both main terminals are accessible for wiring.

The 2S (6V) module is built the same way with two cells, giving a `B+` and `B-` terminal.

3.2. *Attaching the BMS*

Important: Connect the balance/sampling wires before connecting any power wires to the BMS. The BMS uses these to verify cell voltages before enabling its output.

Step 1 — Balance port (sampling wires). The balance connector has one wire per cell junction. Connect them in order from the most negative point of the pack to the most positive:

1. `B0` to the main negative of the pack (the negative terminal of cell 1).
2. `B1` to the junction between cell 1 and cell 2.
3. `B2` to the junction between cell 2 and cell 3.
4. `B3` to the junction between cell 3 and cell 4.
5. `B4` to the main positive of the pack (the positive terminal of cell 4).

For the 2S module, connect only `B0`, `B1`, and `B2`.

Step 2 — Main negative (`B-`). Run a heavy gauge wire from the main negative terminal of the battery pack (`B-`) to the `B-` terminal on the BMS. This is the negative power input to the BMS.

Step 3 — Do not connect `P-` yet. Leave the `P-` terminal disconnected until the rest of the circuit is wired.

3.3. *Wiring the positive path*

Run a heavy gauge wire from the main positive terminal of the battery pack (`B+`) to the input lead of the inline fuse holder. Insert a 20 A ATO blade fuse into the holder. From the output lead of the fuse holder, run a wire to one blade terminal of the TX1 WH2 circuit breaker. The other blade terminal of the TX1 WH2 is the positive output of the module — connect this to the center pin of the N-type output connector.

3.4. *Completing the negative path*

Run a wire from the `P-` terminal of the BMS to the shield of the N-type output connector. This completes the circuit: `B-` enters the BMS, and the protected `P-` output exits to the load via the connector shield.

The full circuit for each module is shown in Figure 1.

3.5. *Final check before connecting a load*

Before connecting any load, verify the following:

1. All balance wires are seated in the BMS balance port.
2. The `B-` wire is connected to the BMS `B-` terminal.
3. The `P-` wire is connected to the BMS `P-` terminal and to the N-type shield.
4. The positive path runs: battery `B+` → inline fuse holder → TX1 WH2 → N-type center pin.
5. The TX1 WH2 breaker is in the OFF position before connecting a load.
6. Measure the voltage between the N-type center pin and shield with a multimeter. It should read the full pack voltage: 12.8–14.6 V for the 4S module (nominal 12.8 V, fully charged 14.4–14.6 V) or 6.4–7.3 V for the 2S module (nominal 6.4 V, fully charged 7.2–7.3 V). If it reads zero with the breaker OFF, that is correct — flip the breaker ON to verify.

4. OPERATING THE SYSTEM

To power on a module, flip its TX1 WH2 breaker to the ON position. To power off, flip it to OFF. The two modules are completely independent and can be switched on or off separately.

If the BMS trips due to a fault (overcurrent, cell overvoltage, or cell undervoltage), the `P-` output opens and the load loses power. The breaker handle will remain in the ON position — this is not a breaker trip. Identify and resolve the fault, then power-cycle the BMS by briefly disconnecting and reconnecting `B-`.

5. CHARGING

5.1. *Charging adapters*

The 12V module output is an N-type **male** connector; the 6V module output is an SMA **male** connector. The polarity convention is the same throughout: center pin = positive, outer shield = negative.

12V module wall charger adapter (N-type male to Anderson Powerpole). Built from a short coaxial jumper with an N-type **male** connector on one end (Figure 5). The center conductor and braid shield are separated and each soldered to a short lead terminated with an Anderson Powerpole (PP) contact: center conductor (positive) to the red contact, braid shield (negative) to the black contact. The two contacts are seated into a single PP housing matching the bench charger standard. Because both the battery output and this adapter terminate in male N-type connectors, a female-to-female N-type barrel adapter is required to mate them.

12V module solar adapter (N-type male to bare wire). Same construction as above but the far end terminates in bare 12–14 AWG wire for direct connection to the MPPT charge controller's (e.g., Genasun GV-10 or equivalent) battery screw terminals: center conductor (positive) to battery +, braid shield (negative) to battery −. Also requires the female-to-female N-type barrel to mate to the battery.

6V module adapter (SMA female to bare wire). A short SMA **female** solder connector with red and black 12–14 AWG wire leads soldered directly to the center pin (positive) and outer shell (negative) (Figure 6). Mates directly to the battery SMA male output with no barrel required. The far end connects to the C65 balance charger output leads for wall charging, or to the C65 output leads powered via an MPPT charge controller for solar charging. No N-type adapter is used in the 6V charging chain.

Connector chain for charging:

- **12V (4S) module, wall charger:** battery N-type (male) → N-type female-to-female barrel → N-type-to-Anderson adapter (male N-type) → Anderson PP → charger.
- **12V (4S) module, solar:** battery N-type (male) → N-type female-to-female barrel → N-type-to-bare-wire adapter (male N-type) → MPPT charge controller battery terminals.
- **6V (2S) module, wall charger:** battery SMA (male) → SMA female adapter → C65 output leads → C65 (set to LiFe 2S, powered via bench supply 11–18 V).

- **6V (2S) module, solar:** solar panel → MPPT charge controller (e.g., Genasun GV-10 or equivalent) solar input → controller DC output → C65 DC input (11–18 V) → C65 output leads → SMA female adapter → battery SMA (male). The MPPT controller powers the C65, which handles the correct 2S $LiFePO_4$ charge profile. Ensure the controller output voltage is within the C65 DC input range (11–18 V).

Caution: the coaxial jumper used in the 12V N-type-to-Anderson adapter (RG-223/U) has a 19 AWG center conductor and is rated for several amps continuous in free air, well below its mechanical/dielectric limit if run at the full 20 A breaker rating for extended periods. The adapter as built has been tested at 20 A charge current for charge sessions of 1+ hours at ∼65°F ambient with the cable remaining warm but not hot, and has not shown signs of thermal degradation. The adapter should not be left charging unattended for multi-hour sessions until it is rebuilt with the N-type connector terminated directly to 12–14 AWG stranded wire rather than coaxial cable, which removes the thermal concern entirely. This rebuild is recommended as a future improvement but is not required for normal operation. The 6V SMA adapter and the 12V solar adapter, both built directly with 12–14 AWG bare wire, do not have this concern.

5.2. *Charger voltage requirements*

Each module requires a charger set to the correct maximum charge voltage for its cell count. Using the wrong charger voltage risks overcharging the cells; the BMS will trip to protect them, but relying on the BMS as the sole safeguard against an incorrect charger is bad practice.

- **12V (4S) module:** charger must be set to 14.4–14.6 V maximum.
- **6V (2S) module:** charger must be set to 7.2–7.3 V maximum.

Do not use the 14.6 V (4S) charger on the 6V (2S) module under any circumstances. The Heltec BMS on the 2S module will trip on overvoltage, but the cells may still be exposed to elevated voltage briefly before the BMS responds. Always verify charger voltage setting before connecting to a module.

5.3. *Charging via wall/bench charger*

12V (4S) module — wall charger:

1. Confirm the TX1 WH2 breaker is in the OFF position.
2. Connect the N-type female-to-female barrel to the battery N-type male output.
3. Connect the N-type male end of the charging adapter to the other end of the barrel.
4. Connect the Anderson PP end to the charger output.
5. Flip the TX1 WH2 breaker to ON.
6. Turn on the charger and confirm it begins its charge cycle.
7. Periodically check the temperature of the adapter cable and breaker terminals by touch. The cable should remain warm, not hot. If any component becomes hot to the touch, turn off the charger and breaker immediately and inspect before resuming.
8. Pack voltage will change very little over the first hour of charging ($LiFePO_4$ has a flat charge curve). Voltage will rise more rapidly only as the pack approaches full charge (14.4–14.6 V).
9. Once charging is complete, turn off the charger first, then flip the TX1 WH2 breaker to OFF before disconnecting the adapter.

6V (2S) module — C65 balance charger:

1. Set the C65 to LiFe chemistry, 2S cell count. Do not use LiPo mode.
2. Connect the C65 output leads to the SMA female adapter (red lead to center pin, black lead to outer shell).

3. Connect the SMA female adapter to the battery SMA male output.
4. Power the C65 via its DC input (11–18 V) from a bench supply or 12V source.
5. Start the charge cycle on the C65 and confirm it reads 2S cell voltages correctly via the balance port if connected.
6. Once charging is complete, disconnect in reverse order.

5.4. *Charging via solar (MPPT charge controller)*

12V (4S) module — MPPT charge controller (14.2V $LiFePO_4$, e.g., Genasun GV-10):

1. Wire the solar panel leads to the MPPT controller's solar/PV input terminals, observing correct polarity.
2. With the TX1 WH2 breaker OFF, connect the N-type female-to-female barrel to the battery N-type male output.
3. Connect the N-type male end of the solar adapter to the other end of the barrel.
4. Connect the bare-wire end of the adapter to the controller's battery-output terminals: positive (center conductor) to battery +, negative (shield) to battery −.
5. Flip the TX1 WH2 breaker to ON.
6. The MPPT controller will regulate charge current based on available sunlight; thermal loading on the adapter is lower than wall charging.
7. Full charge indicated by 14.4–14.6 V at the battery terminals.

6V (2S) module — C65 powered by MPPT charge controller:

1. Wire the solar panel leads to the MPPT controller's solar/PV input terminals.
2. Connect the controller's battery-output terminals to the C65 DC input (11–18 V): controller positive to C65 DC+, controller negative to C65 DC−.
3. Set the C65 to LiFe chemistry, 2S cell count.
4. Connect the C65 output leads to the SMA female adapter and connect to the battery SMA male output.
5. Start the C65 charge cycle. The MPPT controller powers the C65, which handles the correct 2S $LiFePO_4$ charge profile (7.2–7.3 V).
6. Full charge indicated by 7.2–7.3 V at the battery terminals.

Field charging for both modules is expected to occur away from the EIGSEP instrument to avoid any possibility of conducted or radiated RFI from the MPPT controller's switching circuitry coupling into the receiver chain.

5.5. *Runtime and charge time analysis*

Table 1 summarizes expected runtime and charge time for each module under typical EIGSEP loads, assuming 20 A maximum charge current (set by the inline fuse and TX1 WH2 breaker rating) and 80–90% usable capacity for $LiFePO_4$.

The 12V module is the operational bottleneck, with a worst-case runtime of ∼23 hr against a charge time of ∼12 hr. Two 12V modules in rotation — one charging while the other is deployed — therefore comfortably support continuous operation: a fully depleted module finishes charging in less time than the other module takes to deplete, even in the worst-case load scenario. The same holds for the 6V module with substantial margin.

Table 1. Runtime and charge time by module

Module	Typical load	Runtime	Charge time (20 A)
12V (4S, 230 Ah)	5–10 A (stepper motors)	23–46 hr	∼12 hr
6V (2S, 100 Ah)	<1 A (IMU/electronics)	>90 hr	∼5 hr

6. FIGURES

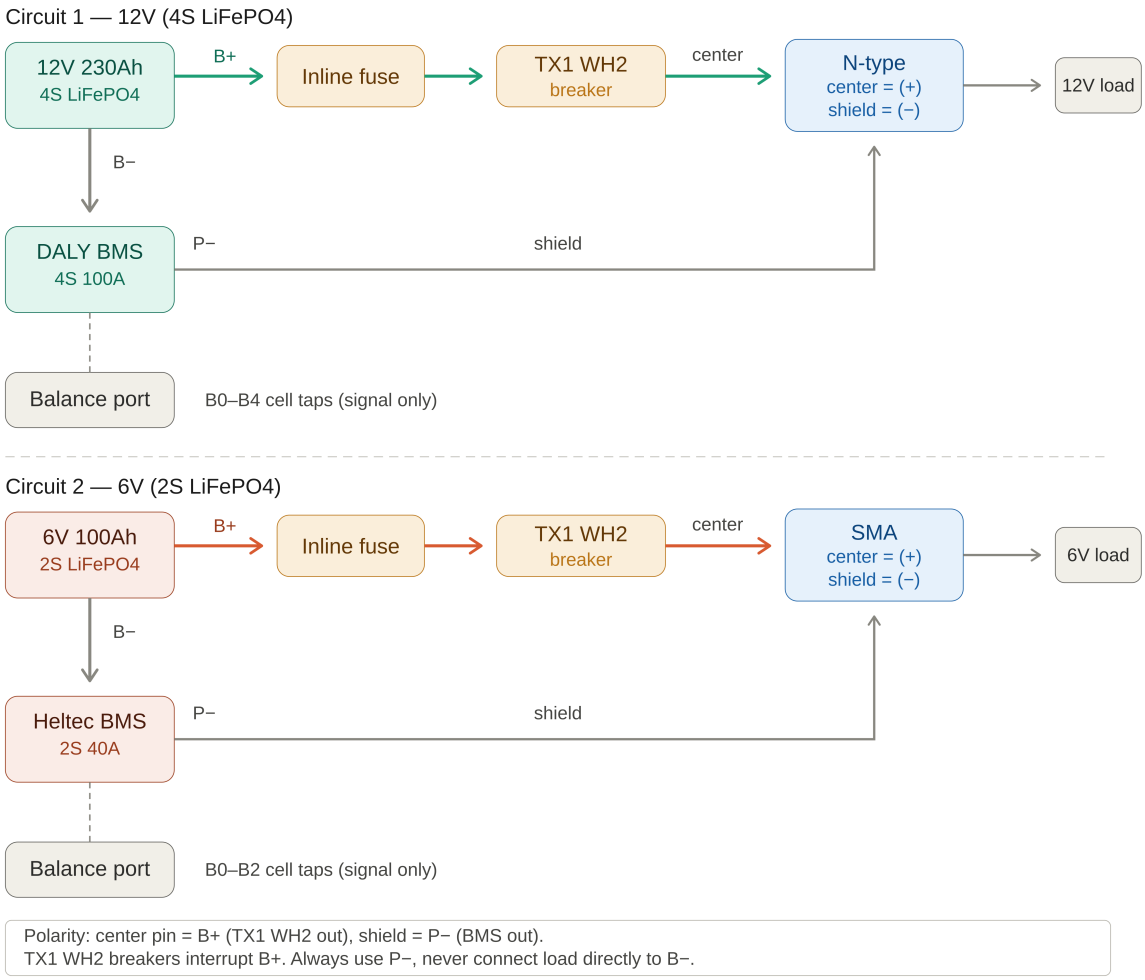


Figure 1. Full wiring schematic for both the 12V (4S) and 6V (2S) circuits. Each circuit is independent and has its own inline fuse holder. Positive path (green/red): battery `B+` → inline fuse holder → TX1 WH2 breaker → output connector center pin → load. Negative path: battery `B-` → BMS → `P-` → output connector shield → load. The balance port (dashed line) carries cell-voltage signals only and is not part of the power path. The 12V (4S) module output connector is N-type male; the 6V (2S) module output connector is SMA male.

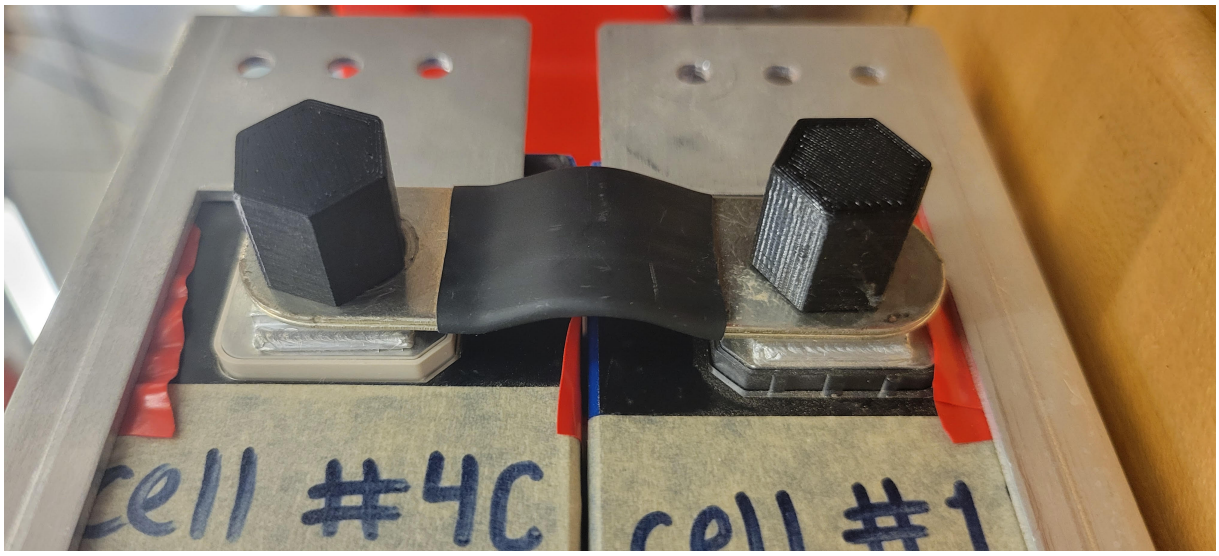


Figure 2. Two $LiFePO_4$ cells connected in series using a shunt. The negative terminal of the lower cell connects to the positive terminal of the upper cell. Repeating this connection across all cells in the module produces the full pack voltage.

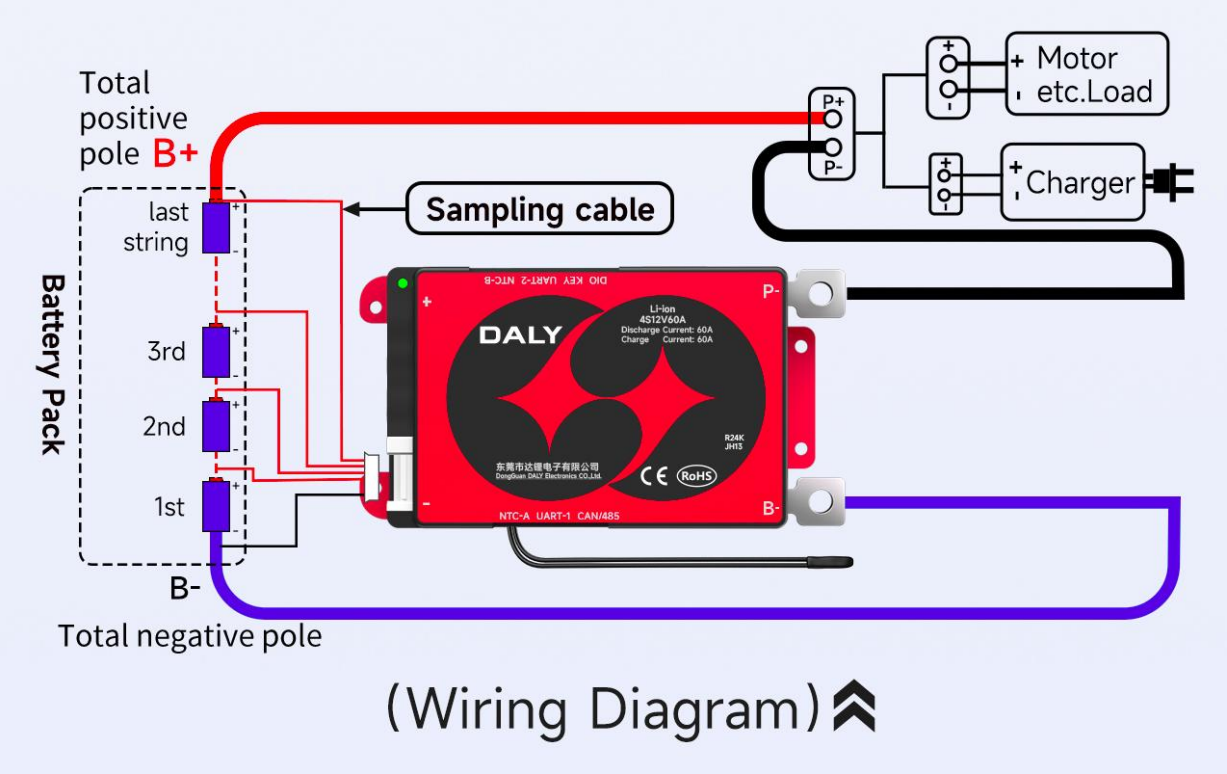


Figure 3. DALY BMS wiring diagram for the 4S (12V) module, showing the balance port connections (B0–B4) and the main power terminals (`B-` input and `P-` output). The 2S module BMS connects identically but uses only B0–B2.

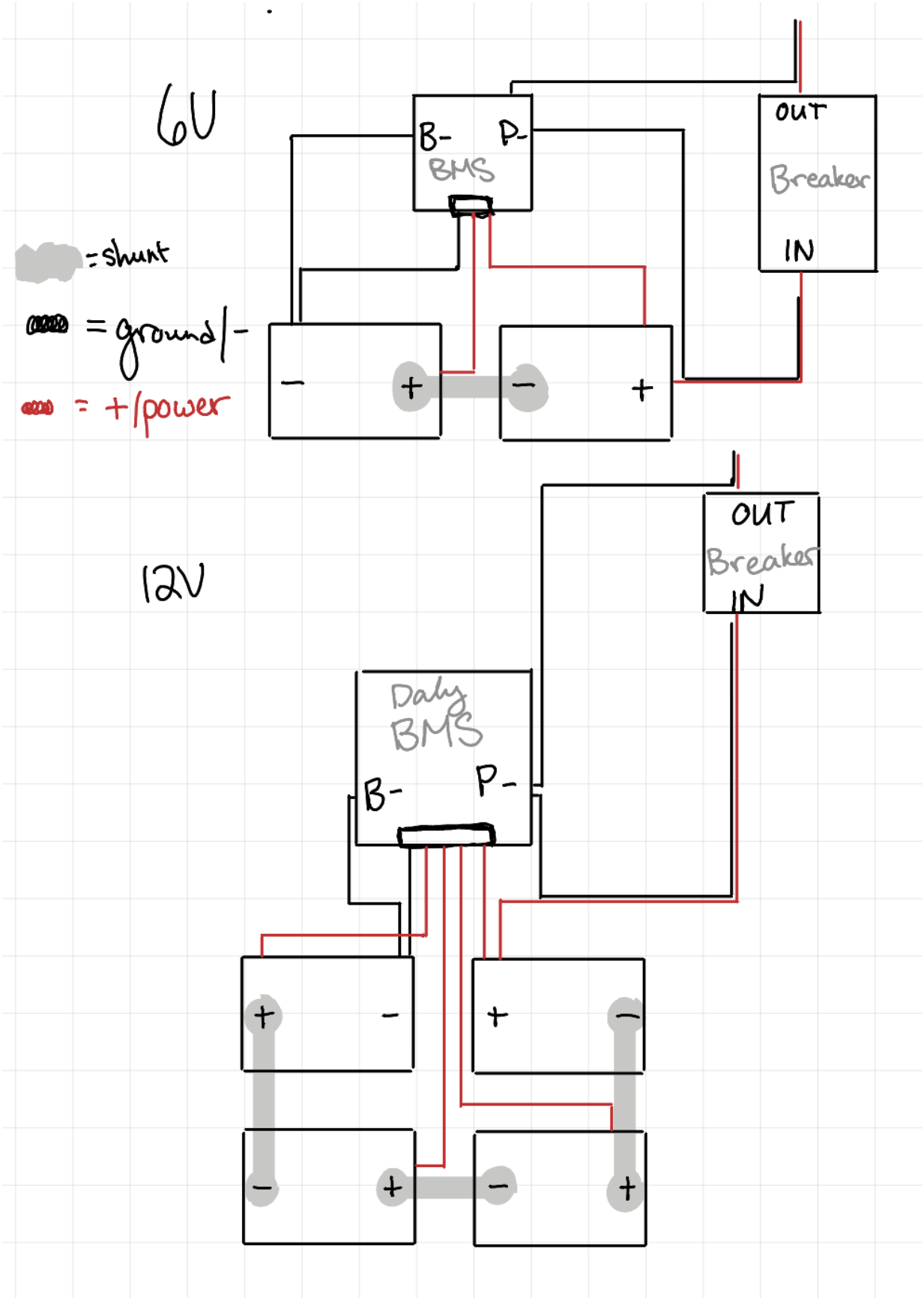


Figure 4. Physical layout of both modules showing cell arrangement, BMS placement, and terminal locations for the 12V (4S) and 6V (2S) packs.



Figure 5. N-type male to Anderson Powerpole charging adapter for the 12V (4S) module. Center conductor (positive) terminates to the red Anderson PP contact; braid shield (negative) terminates to the black contact. A female-to-female N-type barrel is used to mate this adapter to the battery's N-type male output connector.

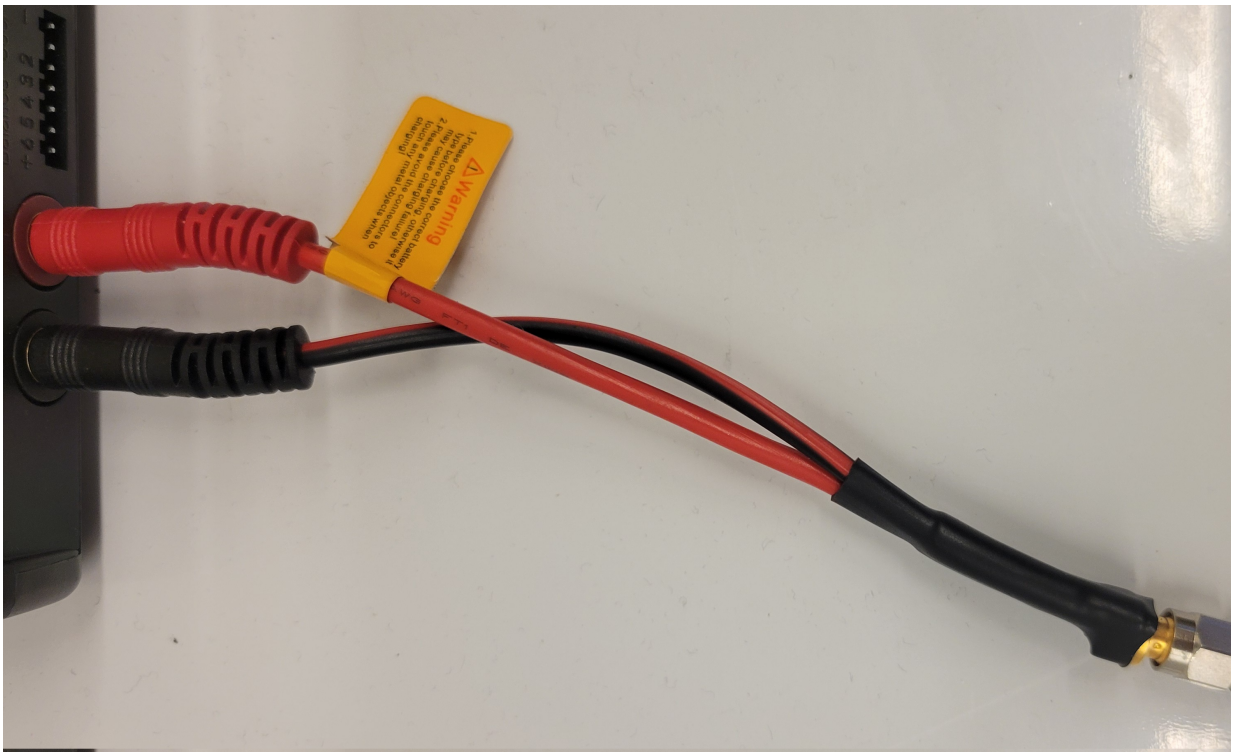


Figure 6. SMA female to bare wire charging adapter for the 6V (2S) module. Red wire soldered to the SMA center pin (positive) and black wire to the outer shell (negative). Far end connects to the C65 balance charger output leads.

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